Forecasting FTSE 100 Monthly Returns: S&P 500, UK Bond Yields, and GBP/USD Effects

The aim of this study is to evaluate whether the past trends in S&P 500, UK 10-year government bond yield, and GBP/USD exchange rates can predict the future returns of FTSE 100 stock index. The dependent variable is the FTSE_RET while SP500_RET, DUK10Y, and GBPUSD_RET are the independent variables.

I converted the index and exchange-rate levels into monthly percent returns and the bond yield into monthly changes to make them stationary. As confirmed by the ADF test, FTSE_RET, SP500_RET, DUK10Y, and GBPUSD_RET are stationary. From the OLS regression analysis, it was revealed that S&P 500 return has a positive effect on FTSE 100 returns.

Specifically, a 1% increase in S&P 500 monthly returns is associated with approximately a 0.746% increase in FTSE 100 monthly returns. GBP/USD returns are statistically significant but have a negative impact while the UK 10-year bond-yield changes are insignificant. The benchmark model explains about 63.3% of FTSE return variation.

The VAR and Granger causality results show limited forecasting power. Past S&P 500 returns provide weak forecasting evidence at the 10% level, while past UK bond-yield changes and past GBP/USD returns do not significantly forecast FTSE returns. The Johansen test shows no cointegration, so a VEC model is not appropriate.

The UK market moves closely with the U.S. market in the same month, but last month's U.S., bond or currency movements don't reliably predict next month's FTSE return.